Cloud Computing & DevOps - Task 3

CI/CD Automation for Dockerized Applications Using GitHub Actions

Objective

In **Task-2**, you learned how to:

- Create Docker images
- Run containers locally
- Deploy Docker containers manually on a Cloud Virtual Machine (AWS EC2)
- Access applications using a public IP
- Understand **manual deployment as the foundation before CI/CD** Cloud Computing & DevOps– Task ...

In **Task-3**, you will **automate the same workflow** using a **CI/CD pipeline**, so that:

- Docker images are built automatically
- Human error is reduced
- Deployments become repeatable and consistent

This task introduces **real DevOps automation workflows** used after manual deployment mastery.

Why This Task Matters (Industry Context)

According to your Task-2 document:

> "Manual deployment is the foundation before CI/CD"
> (page 2 – Concepts You Will Learn) Cloud Computing & DevOps– Task ...

In real companies:

- First, engineers learn manual deployment
- Then, they automate the same steps
- CI/CD is never taught without Docker and VM knowledge

This task reflects **exactly that professional progression**.

What You Will Build

You will build a **CI/CD pipeline** that:

- Automatically builds a Docker image on every Git push
- Pushes the image to Docker Hub
- Ensures the image is always deployment-ready
- Prepares the application for automated VM deployment in the next task

This task focuses on **CI + Image Delivery**, not VM replacement.

Tools & Technologies Used

- Git & GitHub
- GitHub Actions
- Docker
- Docker Hub (Free Account)
- YAML configuration

All tools are free and beginner-safe.

Prerequisites (Mandatory)

Before starting Task-3, ensure:

- Task-2 is fully completed
- Docker image runs correctly on EC2 manually
- Project is pushed to GitHub
- Docker Hub account is created

These prerequisites are explicitly implied in Task-2 deliverables Cloud Computing & DevOps– Task ... .

Part A: CI/CD Fundamentals

Step 1: Understand CI/CD in Context of Task-2

Manual workflow (Task-2):

1. Write Dockerfile
2. Build Docker image
3. Run container
4. Deploy on EC2

Automated workflow (Task-3):

1. Push code to GitHub
2. CI builds Docker image automatically
3. Image is stored centrally
4. Ready for deployment anytime

Only **steps 2 and 3** are automated here.

Part B: Prepare Repository for CI/CD

Step 2: Repository Structure

Your GitHub repository must contain:

```markdown
project-root/
│
├── index.html
├── Dockerfile
├── README.md
└── .github/
    └── workflows/
        └── docker-ci.yml
```

This is the same project used in Task-2, not a new one.

Step 3: Verify Dockerfile (From Task-2)

```dockerfile
FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80
```

Test locally before automation:

```bash
docker build -t portfolio-website .
docker run -d -p 8080:80 portfolio-website
```

This validation step mirrors Task-2 local testing (page 4) Cloud Computing & DevOps– Task ... .

Part C: Create GitHub Actions CI Pipeline

Step 4: Create Workflow File

Create the file:

```bash
.github/workflows/docker-ci.yml
```

Step 5: CI Pipeline Configuration

```yaml
name: Docker CI Pipeline

on:
  push:
    branches:
      - main

jobs:
  docker-build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Code
        uses: actions/checkout@v3

      - name: Login to Docker Hub
        uses: docker/login-action@v2
        with:
          username: ${{ secrets.DOCKER_USERNAME }}
          password: ${{ secrets.DOCKER_PASSWORD }}

      - name: Build Docker Image
        run: docker build -t ${{ secrets.DOCKER_USERNAME }}/portfolio-website:latest .

      - name: Push Docker Image
        run: docker push ${{ secrets.DOCKER_USERNAME }}/portfolio-website:latest
```

Step 6: Configure GitHub Secrets

In GitHub Repository:

- Settings → Secrets → Actions
- Add:
  - `DOCKER_USERNAME`
  - `DOCKER_PASSWORD`

This follows **security best practices**, not hardcoding credentials.

Part D: Execute CI/CD Pipeline

Step 7: Trigger Automation

```bash
git add .
git commit -m "Added Docker CI pipeline"
git push origin main
```

Step 8: Verify Pipeline Execution

- Open GitHub → Actions tab
- Confirm:
  - Code checkout
  - Docker build success
  - Docker image pushed

Step 9: Validate Image from Docker Hub

On any system (or EC2):

```bash
docker pull yourusername/portfolio-website:latest
docker run -d -p 80:80 yourusername/portfolio-website
```

If the site loads, CI is working correctly.

Expected Output

After completing Task-3:

- Docker image builds automatically
- No manual image creation required
- Image is centrally available
- Deployment consistency is achieved

Key Learning Outcomes

After Task-3, interns will be able to:

- Explain CI/CD clearly
- Automate Docker image builds
- Use GitHub Actions confidently
- Secure credentials properly
- Transition from manual to automated workflows

Deliverables

Mandatory:

1. GitHub repository with CI workflow
2. Screenshot of successful GitHub Actions run
3. Docker Hub image link
4. README explaining CI process

Optional:

- Versioned Docker tags
- CI status badge

1. Core Tools & Accounts (Free)

Git & GitHub

- Git download: https://git-scm.com/downloads
- GitHub account: https://github.com

Docker

- Docker Desktop: https://www.docker.com/products/docker-desktop/

Docker Hub

- Free account: https://hub.docker.com

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2. CI/CD Fundamentals (Must Understand First)

What is CI/CD (Clear Beginner Explanation)

https://www.youtube.com/watch?v=42UP1fxi2SY

CI vs CD Explained with Examples

https://www.youtube.com/watch?v=scEDHsr3APg

DevOps CI/CD Pipeline Explained

https://www.youtube.com/watch?v=OPwU3UWCxhw

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3. GitHub Actions Basics (Core for Task-3)

GitHub Actions Explained from Scratch

https://www.youtube.com/watch?v=R8_veQiYBjI

GitHub Actions Full Beginner Tutorial

https://www.youtube.com/watch?v=TLB5MY9BBa4

GitHub Actions Workflow YAML Explained

https://www.youtube.com/watch?v=yr7n0qf5C3o

4. Docker + CI/CD Integration (Very Important)
Build Docker Image Using GitHub Actions
https://www.youtube.com/watch?v=QeU_5e8oQJo
Docker CI Pipeline with GitHub Actions
https://www.youtube.com/watch?v=VqUj1Qp7p2c
Push Docker Image to Docker Hub Automatically
https://www.youtube.com/watch?v=0K2mYpOeZxU

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5. GitHub Secrets & Security (Mandatory Knowledge)
GitHub Secrets Explained
https://www.youtube.com/watch?v=I9tGdW7F9Kw
Using Secrets in GitHub Actions
https://www.youtube.com/watch?v=0a6jKkQ8R2E
Why Secrets Are Important in CI/CD
https://www.youtube.com/watch?v=8pP3tV0zP5E

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6. Dockerfile Best Practices (From Task-2 to Task-3)
Dockerfile Explained Line by Line
https://www.youtube.com/watch?v=Gjnup-PuquQ
Docker Image vs Container Explained
https://www.youtube.com/watch?v=8fi7uSYlOdc
Docker Build and Run Basics
https://www.youtube.com/watch?v=pTFZFxd4hOI

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7. Hands-On End-to-End Tutorials (Task-Aligned)
CI/CD Pipeline for Docker App (Beginner)
https://www.youtube.com/watch?v=9kZ4G6k9P3o
Static Website CI/CD with Docker + GitHub Actions
https://www.youtube.com/watch?v=KJq3eG4pF8k

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8. Official Documentation (Free & Reliable)
GitHub Actions Documentation
https://docs.github.com/en/actions
GitHub Actions Workflow Syntax
https://docs.github.com/en/actions/using-workflows/workflow-syntax-for-github-actions
Docker Official Documentation
https://docs.docker.com/
Docker Login GitHub Action
https://github.com/docker/login-action

9. Practice & Reference Repositories
Sample GitHub Actions Docker CI Repositories
https://github.com/search?q=github+actions+docker+ci

Docker CI/CD Examples
https://github.com/search?q=docker+github+actions+ci

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10. Common Errors & Troubleshooting (Very Useful)

Fix Docker Push Permission Denied
https://www.youtube.com/watch?v=3jP0T1P3Z8A
GitHub Actions Build Failed Fixes
https://www.youtube.com/watch?v=7FJ6mK9Z5Zg
Docker Login Issues in CI
https://www.youtube.com/watch?v=5Z3t8JYQ1ZQ

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11. Community Support (Free Help)

- GitHub Actions Discussions
  https://github.com/orgs/community/discussions
- Stack Overflow (GitHub Actions tag)
  https://stackoverflow.com/questions/tagged/github-actions
- Docker Community Forums
  https://forums.docker.com/
- DevOps Reddit
  https://www.reddit.com/r/devops/

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12. Recommended Learning Path for Task-3

Day 1
- CI/CD concepts
- GitHub Actions basics

Day 2
- Docker build and push automation

Day 3
- Secrets management
- Pipeline debugging

Day 4
- End-to-end CI execution
- Image validation

Day 5
- Documentation
- Screenshots and submission

13. Essentials Checklist for Task-3

You must clearly understand:
- CI/CD purpose and flow
- GitHub Actions workflow structure
- Docker build and push automation
- Secure secrets handling
- Pipeline monitoring and logs
- Difference between manual and automated workflows